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"AI-Powered Virtual Teaching Assistant "

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CERTIFICATE

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DECLARATION

We, **AKARSH PR (23BTRCL097)**, **ANANT PRADEEP NAIR (23BTRCL029)**, **AUSTIN (23BTRCL033)**, **BHAVAN ADHITYAN J (23BTRCL099)**, **BHALAMUKUNTHAN VS (23BTRCL168)**, **DHARANEESH R (23BTRCL039)**, **DIWAKAR S (23BTRCL182)** students of IV Semester B.Tech in **Computer Science and Engineering (Artificial Intelligence and Machine Learning)**, at School of Engineering & Technology, Faculty of Engineering & Technology, **JAIN (Deemed-to-be University)**, hereby declare that the PCL work titled **"AI-Powered Virtual Teaching Assistant"** has been carried out by us and submitted in partial fulfillment for the award of the degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2026–2027**.

Further, the matter presented in this work has not been submitted previously by anybody for the award of any degree or diploma to any other University, to the best of our knowledge and belief.

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Contents

1 Introduction **8**

1.1 Motivation and Background 8

1.2 Aim and Objective 9

2 Literature Survey **10**

2.1 Literature Survey 10

2.1.1 Literature Review 10

2.1.2 Inferences Drawn from Literature Review 13

3 Problem Statement and Proposed System **15**

3.1 Problem Statement 15

3.1.1 Lack of Real-Time Interaction 15

3.1.2 Limited Contextual Understanding 15

3.1.3 Absence of Personalized Learning 16

3.1.4 Single-Mode Communication 16

3.1.5 Low User Engagement 17

3.1.6 Fragmented Integration of Technologies 17

3.2 Proposed System 18

3.2.1 System Modeling 18

3.2.2 State-Based Interaction Modelling 19

3.2.3 Multimodal Response Generation 20

3.3 Architectural Modeling 20

3.3.1 Layer 1: User Interaction and Engagement 20

3.3.2 Layer 2: Signal Processing and Tokenization 20

3.3.3 Layer 3: AI Intelligence and Context Management 21
3.3.4 Layer 4: Generative Multimodal Synthesis 21
3.3.5 Layer 5: Avatar Interaction and Behavioural Simulation 21
3.4 Adaptive Learning Mechanism 23

4 Implementation And Testing **24**
4.1 Importing Necessary Packages 24
4.2 Importing a Model 24
4.3 Fine Tuning and Optimizing the Model 25
4.4 Making An API For the Model Using Hugging Face 26
4.5 Project Structure 27

5 Testing and Evaluation **28**
5.1 Prototype Implementation 28
5.2 Functional Testing and Observed Behavior 28
5.2.1 Behavior Observed 28
5.3 Identified Limitations 29
5.4 Summary of Findings 29

6 Conclusion **31**

List of Figures

3.1 Architecture Diagram 22

4.1 Importing packages 24
4.2 Pretrained Model import 24
4.3 bnb config 25
4.4 SFT Training 25
4.5 Output 26
4.6 Creating model API 26
4.7 Project Structure 27

ABSTRACT

Rapid development in AI technologies has enabled the emergence of different approaches to the creation of intelligent learning environments. Regrettably, many e-learning platforms available today lack the necessary level of interactivity, provide no personalized assistance to the learner, and fail to engage the audience efficiently. This paper outlines the process of designing a virtual tutor or conversational agent capable of interacting via speech recognition and text input and applying AI-based approaches for text, voice, and image generation powered by cloud services. In this regard, the suggested framework depends on the use of AI-based services offered by cloud service providers to ensure multilateral learning through text, voice, and image generation. A prototype of the virtual tutor has been designed and tested for functionality. As shown during testing, the proposed system is capable of processing requests either by voice or text, keeping record of the context, generating responses both by voice and text, as well as creating images according to user requests. To coordinate visual and audio feedbacks, an avatar with an authentic appearance is used. Despite being technologically viable, the discussed approach requires further empirical evaluation and personalization.

1 Introduction

Artificial Intelligence (AI) has rapidly evolved in recent years and is now influencing many areas of everyday life, including education. With the growth of high-speed internet, affordable smartphones, and cloud-based platforms, access to educational content has increased significantly. Students from different parts of the world can now access learning materials that were previously available only in well-equipped institutions. However, despite this increase in accessibility, many digital learning systems still struggle to provide an effective learning experience. One major challenge is that most online platforms cannot replicate the level of interaction, guidance, and personalized support that students receive in traditional classroom environments.

Most current e-learning platforms follow a passive learning approach. They mainly rely on pre-recorded video lectures, static reading materials, and automated quizzes that provide only basic feedback. When a student does not understand a concept while watching a lecture, the platform usually does not provide an easy way to ask questions or get immediate clarification. As a result, students often need to pause the lecture and search for explanations from external sources, or continue learning without fully understanding the topic. This lack of timely support can negatively affect the learning process. Studies have shown that students who cannot receive quick and relevant help when facing difficulties are more likely to lose interest and drop out. This is one of the key reasons why many large online courses report very low completion rates, even when thousands of students initially enroll.

1.1 Motivation and Background

Digital education is one of the main ways many people use to learn, By looking into the current implementations we can clearly see that the platforms were less effective despite being one of the greatest revolutions in the Education sector.

This is mainly due to the disconnect the learners feel when they are taking on a online course or a lecture, Many find it difficult as their pace of processing information varies from

an average learner for which all the online platforms or designed for.

This created friction and ultimately caused many to abandon the idea of online learning. To solve this we are proposing an AI teacher who can teach the learners at their own pace through extensive personalization and options, so learners not only can engage better, but also end up learning new things which they might actually benefit from.

1.2 Aim and Objective

Aim:

The aim of this project is to develop an AI-powered Virtual Teaching Assistant that enhances digital learning by providing interactive, personalized, and real-time educational support to learners.

Objectives:

- To design an intelligent system capable of understanding student queries using Natural Language Processing.
- To implement real-time interaction through both text and voice-based communication.
- To develop a context-aware system that can maintain conversation continuity and provide relevant responses.
- To introduce personalization mechanisms that adapt learning content based on the learner's pace and feedback.
- To integrate multimodal interaction such as voice responses and visual explanations for improved engagement.
- To improve student motivation and learning efficiency through an interactive AI-based teaching assistant.

2 Literature Survey

2.1 Literature Survey

A literature survey helps in understanding the existing research related to a particular problem and provides the foundation for developing new solutions. In the field of digital education, several technologies have been explored to improve learning experiences. These technologies include intelligent tutoring systems, conversational artificial intelligence, multimedia learning environments, speech-based interaction systems, and avatar-based educational interfaces. Each of these research areas contributes valuable insights into how digital learning platforms can be made more interactive and effective.

With the rapid development of artificial intelligence, modern educational systems have started integrating AI-based solutions to assist learners. Traditional online learning platforms primarily rely on pre-recorded lectures, reading materials, and automated assessments. Although these resources improve accessibility to education, they often lack real-time interaction and personalized learning support. As a result, many students experience difficulties when they encounter complex concepts that require immediate clarification.

Researchers have attempted to address these limitations through the development of intelligent tutoring systems and conversational learning agents. These systems aim to simulate the role of a human instructor by providing guidance, answering questions, and adapting the learning process based on the learner's progress. The following sections review the most relevant research contributions that have influenced the design of the proposed AI-powered virtual teaching assistant.

2.1.1 Literature Review

One of the earliest research areas related to this project is the development of Intelligent Tutoring Systems (ITS). Intelligent tutoring systems were designed to provide personalized instruction using computer-based systems. Early research suggested that computers could analyze a student's knowledge and provide tailored educational guidance. These systems typically consisted of several components including a domain model representing subject knowledge, a student model tracking the learner's progress, and a pedagogical model that

determined the instructional strategy.

Several successful implementations of intelligent tutoring systems were developed for subjects such as mathematics and programming. These systems demonstrated that personalized instruction could significantly improve student learning outcomes. However, early tutoring systems were limited to well-structured subjects and required significant manual design of rules and knowledge models.

To improve the interaction between learners and tutoring systems, researchers introduced conversational tutoring systems that used natural language dialogue. One well-known example is AutoTutor, which engaged students in question-and-answer conversations while teaching complex subjects. Instead of directly giving answers, the system guided learners through hints, explanations, and follow-up questions. Studies showed that such conversational approaches could improve student understanding compared to passive learning methods.

However, early conversational tutoring systems were restricted by the limitations of traditional natural language processing techniques. These systems relied on keyword matching and predefined response patterns, which limited their ability to understand diverse student inputs. As a result, they could not fully replicate the flexibility of human tutors.

A major breakthrough in conversational AI occurred with the introduction of transformer-based language models. The transformer architecture enabled machines to process language more effectively by understanding contextual relationships between words. This advancement led to the development of powerful language models such as BERT and GPT, which significantly improved natural language understanding and text generation capabilities.

These models can interpret user queries, maintain conversational context, and generate coherent responses. As a result, they provide a strong foundation for building AI-based tutoring systems capable of supporting interactive educational dialogue. The ability of large language models to explain concepts, answer questions, and provide contextual responses has opened new possibilities for AI-assisted learning environments.

Another important area of research influencing this project is multimedia learning. According to the Cognitive Theory of Multimedia Learning, students learn more effectively when information is presented through multiple channels such as text, visuals, and audio.

The theory explains that the human brain processes visual and verbal information separately, and combining these channels can enhance understanding.

Research in multimedia learning has demonstrated that combining explanations with diagrams, illustrations, or animations can significantly improve comprehension. For example, complex scientific or technical topics are easier to understand when visual representations accompany textual explanations. These findings support the integration of multimodal outputs in educational systems.

Cognitive Load Theory further emphasizes the importance of instructional design in learning environments. The theory states that learners have limited cognitive capacity, and poorly structured information can overload the learner's working memory. Effective instructional systems should therefore present information in manageable segments and reduce unnecessary complexity.

Avatar-based pedagogical agents represent another important research direction. These agents are animated characters designed to interact with learners in digital environments. Research has shown that learners often respond positively to animated teaching agents because they create a sense of social interaction. This phenomenon, known as the persona effect, suggests that the presence of an animated character can improve learner motivation and engagement.

Studies involving animated agents demonstrated that learners reported higher levels of satisfaction and interest when interacting with systems that included a visual instructor. Although the instructional content remained the same, the presence of an animated guide improved the overall learning experience.

Speech-based interaction has also gained significant attention in educational technology. Voice interaction allows learners to communicate with AI systems in a natural and intuitive manner. Research comparing voice-based and text-based tutoring systems found that voice interaction often makes conversations more engaging and easier for students.

Advances in automatic speech recognition have significantly improved the reliability of voice-based systems. Modern speech recognition models can accurately convert spoken language into text, enabling AI systems to process voice queries effectively. Similarly, text-to-

speech technologies allow systems to generate natural-sounding audio responses.

Generative artificial intelligence represents one of the most recent developments in educational technology. Generative AI models are capable of producing new content such as text explanations, images, and examples. In educational contexts, these capabilities can be used to generate explanations, visual aids, and personalized learning materials.

For example, generative image models can create diagrams or visual illustrations based on textual descriptions. This allows educational systems to produce visual explanations dynamically rather than relying solely on pre-created educational materials.

2.1.2 Inferences Drawn from Literature Review

The literature survey highlights several key findings that influence the design of the proposed AI-powered virtual teaching assistant.

First, interactive dialogue is essential for effective digital learning. Research on intelligent tutoring systems demonstrates that students benefit from systems that allow them to ask questions and receive immediate responses rather than relying only on static educational content.

Second, multimodal learning significantly improves understanding. Educational studies consistently show that combining text, visuals, and audio explanations leads to better learning outcomes compared to single-mode content delivery.

Third, avatar-based interaction can enhance learner engagement. The presence of animated teaching agents helps create a more interactive learning environment and motivates learners to continue using the system.

Fourth, voice interaction improves communication between learners and AI systems. Speech-based interaction allows users to interact more naturally and reduces the effort required to type complex questions.

Fifth, generative artificial intelligence enables the dynamic creation of educational content. AI models can generate explanations, examples, and visual representations tailored to the learner's needs.

Sixth, maintaining conversational context is important for meaningful educational dialogue. AI tutoring systems must be able to remember previous interactions and maintain continuity in conversations.

Finally, integrating these technologies into a unified system is necessary for creating an effective digital learning platform. Existing solutions often implement these technologies separately, but the proposed system combines conversational AI, multimodal learning, avatar interaction, voice communication, and adaptive learning mechanisms into a single integrated architecture.

Sr. No	**Paper / Topic**	**Author**	**Year**	**Methods / Approach**
1	Intelligent Tutoring Systems for Personalized Learning	Carbonell et al.	1970	Early AI-based tutoring systems for personalized instruction
2	AutoTutor: Conversational Intelligent Tutoring System	Graesser et al.	2001	Natural language dialogue for interactive tutoring
3	Attention Is All You Need	Vaswani et al.	2017	Transformer architecture for language understanding
4	BERT: Bidirectional Encoder Representations from Transformers	Devlin et al.	2019	Deep contextual NLP models for language understanding
5	Language Models are Few-Shot Learners (GPT-3)	Brown et al.	2020	Large-scale generative language models
6	Cognitive Theory of Multimedia Learning	Mayer	2009	Multimodal learning using text, visuals, and audio
7	Animated Pedagogical Agents and the Persona Effect	Lester et al.	1997	Avatar-based interactive learning systems
8	Speech Recognition using Neural Networks	Graves et al.	2013	Deep learning based speech recognition
9	Generative AI Models for Content Generation	Goodfellow et al.	2014	Generative models for creating new data
10	Adaptive Learning in Educational Systems	Brusilovsky	2001	Personalized learning based on student interaction

Table 2.1: Summary of Literature Survey on AI-Based Educational Systems

3 Problem Statement and Proposed System

3.1 Problem Statement

Despite significant advancements in educational technology, existing digital learning platforms continue to face several challenges that limit their effectiveness. These challenges create a gap between traditional classroom learning and online education, giving the need for more advanced and intelligent systems.

3.1.1 Lack of Real-Time Interaction

One of the major limitations of many existing e-learning platforms is the lack of real-time interaction between learners and the system. Most platforms primarily depend on pre-recorded lectures, static study material, and delayed communication methods. In a traditional classroom, students are able to ask questions and receive immediate clarification from instructors. However, digital platforms often fail to provide this level of instant feedback.

The delay in interaction negatively affects the learning process in several ways:

- Reduces the clarity of complex concepts
- Increases students' dependency on external learning resources
- Lowers the overall efficiency of the learning process

Therefore, there is a growing need for intelligent systems capable of providing real-time, interactive responses to students.

3.1.2 Limited Contextual Understanding

Although many modern learning platforms include chatbot-based interfaces, these systems often struggle with understanding the deeper context of user queries. Most of them rely on predefined responses or basic Natural Language Processing techniques that do not fully capture the meaning behind complex questions.

As a result:

- Complex or multi-part queries are not interpreted correctly
- Responses may be incomplete, vague, or irrelevant
- The continuity of conversations is not properly maintained

This limitation significantly reduces the effectiveness of automated learning systems and highlights the need for more context-aware and intelligent solutions.

3.1.3 Absence of Personalized Learning

Students differ in their learning styles, prior knowledge, and pace of understanding. However, many existing digital learning platforms follow a generalized approach to content delivery, presenting the same material in the same manner to all learners.

This one-size-fits-all approach can lead to several issues:

- Slower learners may struggle to keep up with the content
- Advanced learners may not feel sufficiently challenged
- Overall learner engagement and satisfaction may decrease

To address this issue, intelligent systems should be able to adapt dynamically based on user behavior, learning progress, and feedback, enabling a more personalized learning experience.

3.1.4 Single-Mode Communication

Many learning platforms rely heavily on a single mode of communication, typically text or video-based instruction. However, effective learning often requires multiple forms of content presentation to accommodate different learning preferences.

For example:

- Visual learners benefit from diagrams, charts, and images
- Auditory learners understand concepts better through spoken explanations
- Text-oriented learners prefer structured written material

The absence of multimodal interaction can limit both comprehension and long-term knowledge retention.

3.1.5 Low User Engagement

Traditional classroom environments encourage interaction through verbal communication, body language, and emotional connection between instructors and students. In contrast, many digital learning platforms lack these interactive elements, which can lead to lower engagement levels among learners.

This often results in:

- Decreased motivation to participate in learning activities
- Reduced student interaction with the platform
- Lower knowledge retention and completion rates

Incorporating human-like interaction mechanisms, such as avatars and voice-based communication, can significantly improve user engagement in digital learning environments.

3.1.6 Fragmented Integration of Technologies

Another challenge with existing solutions is that many technologies are implemented in isolation rather than as part of a unified system. For example:

- Chatbots are used for text-based interaction
- Multimedia systems are used to deliver educational content
- Voice systems are implemented to improve accessibility

However, these technologies are rarely integrated into a cohesive framework. As a result, their combined potential is not fully utilized.

A unified system that integrates these technologies into a single intelligent learning platform is essential for providing a comprehensive and effective educational experience.

3.2 Proposed System

The rapid growth of digital education has significantly increased the use of online learning platforms. While these platforms have made education more accessible, many of them still struggle to provide an engaging and effective learning experience. Common limitations include the lack of real-time interaction, limited contextual understanding of student queries, and the absence of personalized learning support. To address these challenges, this work proposes an AI-powered Virtual Teaching Assistant that integrates multiple Artificial Intelligence techniques to create a more interactive and immersive learning environment.

The proposed system is designed as an intelligent framework that receives user queries and generates meaningful responses using technologies such as Natural Language Processing (NLP), speech processing, and generative AI models. Unlike traditional systems that mainly depend on static content or predefined responses, the proposed model dynamically generates responses based on the user's input and previous interactions. This allows the system to provide more relevant and context-aware assistance to learners.

From a system modelling perspective, the interaction between the user and the system can be viewed as an input-output transformation process. The user submits a query, which is then processed and converted into a structured form for analysis. Based on this processed information, the system generates an appropriate response by incorporating contextual knowledge and relevant information from its internal knowledge base.

The system is designed to understand user questions, generate context-aware responses, and explain concepts visually when necessary. It communicates with learners through a human-like virtual avatar.

The overall objective of the system is to improve engagement, interactivity, and adaptability in digital learning platforms.

3.2.1 System Modeling

The proposed architecture can be represented as an interactive system that processes user input through several stages such as data collection, language processing, answer generation,

and response delivery.

User input can be represented as a sequence of queries:

$$Q = \{q_1, q_2, \ldots, q_n\}$$

The system processes the query using a Natural Language Processing (NLP) model and produces a response R:

$$R = f(Q, C)$$

where C represents the contextual information obtained from previous user interactions.

For generating meaningful responses, the probability of response generation can be represented as:

$$P(R|Q) = \prod_{i=1}^{n} P(r_i \mid Q, r_1, r_2, \ldots, r_{i-1})$$

This formulation helps the system generate responses sequentially while considering previous outputs.

3.2.2 State-Based Interaction Modelling

The interaction between the user and the system can be described as a sequence of state transitions. Each interaction updates the internal state of the system.

The system state can be defined as:

$$S_t = (Q_t, C_t)$$

where Q_t represents the user query at time t and C_t represents the contextual information at that time.

The next system state can be written as:

$$S_{t+1} = S_t + \Delta S$$

where ΔS represents the change in knowledge or context after processing the user inter-

action.

3.2.3 Multimodal Response Generation

One of the key features of the proposed system is multimodal response generation. The system produces responses using different communication formats such as text, speech, and visuals.

The overall response output can be expressed as:

$$O = T + V + A$$

where:

- T represents the textual response
- V represents visual content such as diagrams or images
- A represents the audio response generated using speech synthesis

3.3 Architectural Modeling

The proposed system architecture is organized into five functional layers. This layered design helps ensure scalability, efficient processing, and real-time responsiveness. Each layer is responsible for a specific task in the interaction pipeline, allowing the system to process user queries effectively and generate meaningful responses.

3.3.1 Layer 1: User Interaction and Engagement

This layer acts as the primary interface between the user and the system. It allows learners to interact with the virtual assistant using both text and voice-based queries. The main goal of this layer is to maximize user engagement by providing a simple and accessible entry point for different types of learners.

3.3.2 Layer 2: Signal Processing and Tokenization

The input processing layer is responsible for converting raw user input into a structured format that can be analyzed by the system.

- **Speech Recognition:** For voice-based queries, the system converts acoustic signals into text using a transformation function $T = f(S)$, where the speech signal is processed into machine-readable text.
- **Semantic Tokenization:** Once the query is received, it is broken down into smaller linguistic units represented as $Q = \{q_1, q_2, ..., q_n\}$. These tokens undergo normalization and filtering to remove irrelevant or noisy information before further processing.

3.3.3 Layer 3: AI Intelligence and Context Management

This layer serves as the core intelligence of the system. It uses Natural Language Processing (NLP) techniques to understand the meaning of user queries and identify the context behind them. A context management module maintains a sliding window of previous interactions, allowing the assistant to remember recent conversations and maintain continuity during discussions.

3.3.4 Layer 4: Generative Multimodal Synthesis

The generation layer integrates advanced AI models to dynamically create responses and learning materials.

- **Generative AI:** The system uses generator functions $G(z)$ to produce dynamic visual explanations and examples, especially for complex scientific or technical topics.
- **Speech Synthesis:** Text-to-speech technology is used to convert generated responses into natural-sounding speech, improving accessibility and creating a more interactive learning experience.

3.3.5 Layer 5: Avatar Interaction and Behavioural Simulation

To improve user engagement in digital learning environments, this layer introduces an embodied virtual instructor. The avatar mimics human-like teaching behaviour and enhances the overall learning experience through several mechanisms:

- **Facial Expressions:** The avatar adjusts its facial expressions based on the sentiment or tone of the generated response.

- **Persona Effect:** The system utilizes the psychological impact of animated pedagogical agents to increase learner motivation and engagement.
- **Lip Synchronization:** The auditory output A is synchronized with the avatar's mouth movements to create a realistic and immersive interaction.

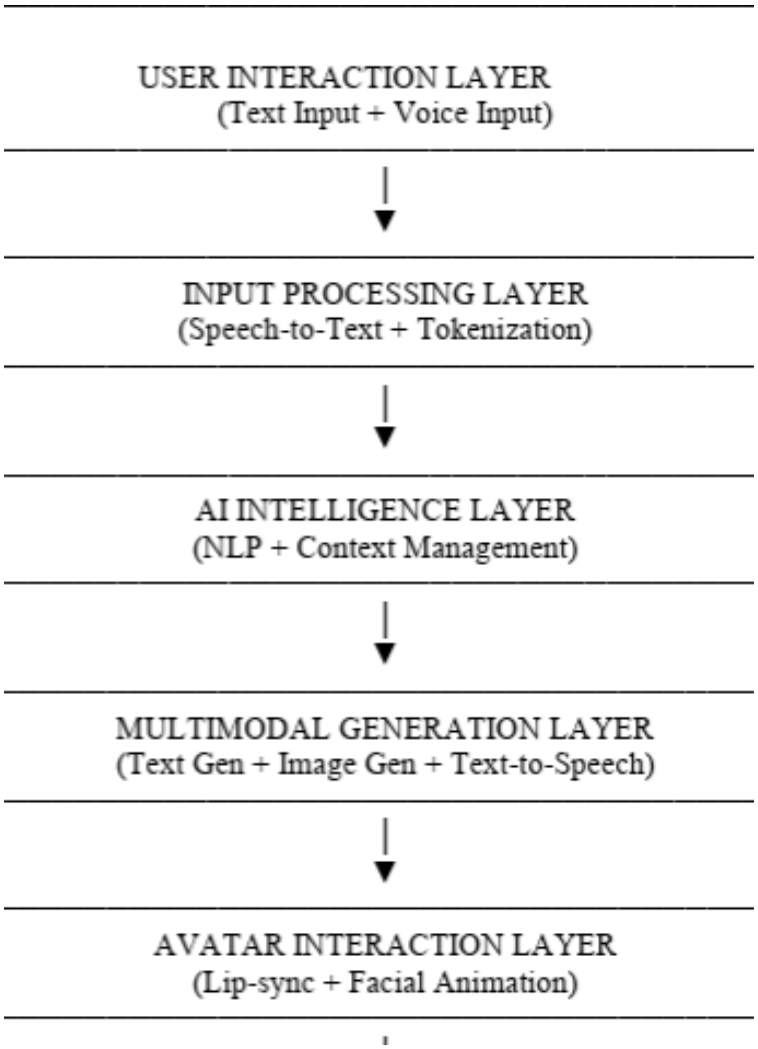

Figure 3.1: Architecture Diagram

3.4 Adaptive Learning Mechanism

Personalization within the proposed system is achieved through an adaptive feedback mechanism that continuously updates the system's understanding of a user's knowledge level. After each interaction, the system evaluates the feedback received from the user and adjusts its internal representation of the learner's knowledge accordingly.

Let U_t represent the user's knowledge state at time t. After each interaction, the system updates this knowledge state based on the feedback received F_t and a predefined learning rate α.

$$U_{t+1} = U_t + \alpha(F_t) \tag{3.1}$$

In this formulation, the learning rate α determines how strongly the feedback influences the update process. A higher value of α allows the system to adapt more quickly, while a smaller value results in more gradual adjustments.

Through this adaptive mechanism, the system gradually evolves from a generalized content delivery platform into a personalized virtual tutor. Over time, it becomes capable of identifying when a learner requires additional explanation, simplified content, or a different mode of communication.

4 Implementation And Testing

4.1 Importing Necessary Packages

```
import torch
from datasets import load_dataset
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig
from peft import LoraConfig, get_peft_model, prepare_model_for_kbit_training
from trl import SFTTrainer, SFTConfig
```

Figure 4.1: Importing packages

4.2 Importing a Model

```
MODEL = "microsoft/Phi-3.5-mini-instruct"
```

Figure 4.2: Pretrained Model import

We are using Microsoft Phi- 3.5 mini LLM model as it's a quite simple model to run locally with only around 3.8B parameters. We can also run any other preferred model such as Llama 3.2 with around 3B parameters or Qwen 2.5 with 7B parameter for more complex topics.

4.3 Fine Tuning and Optimizing the Model

```
bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16,
)

model = AutoModelForCausalLM.from_pretrained(
    MODEL,
    quantization_config=bnb_config,
    device_map="auto",
    trust_remote_code=True
)

tokenizer = AutoTokenizer.from_pretrained(MODEL, trust_remote_code=True)
tokenizer.pad_token = tokenizer.eos_token
```

Figure 4.3: bnb config

We use the BitsAndBytesConfig from hugging face libraries to optimize the model we are using, We load the model in 4 bit mode for efficiency and we are also using the 'nf4' quantization method for greater accuracy.

```
trainer = SFTTrainer(
    model=model,
    train_dataset=dataset,
    args=SFTConfig(
        output_dir="./nexus-teacher",
        num_train_epochs=3,
        per_device_train_batch_size=2,
        gradient_accumulation_steps=4,
        learning_rate=2e-4,
        bf16=True,
        logging_steps=10,
        save_steps=100,
        warmup_steps=50,
        lr_scheduler_type="cosine",
        report_to="none",
    ),
)

trainer.train()
```

Figure 4.4: SFT Training

Fine tuning the model more with other parameters to make the model more efficient and accurate in the answers. These parameters can change according to the model used.

Output

```
Tokenizing train dataset: 100%  138/138 [00:00<00:00, 396.66 examples/s]
WARNING:transformers_modules.microsoft.Phi_hyphen_3_dot_5_hyphen_mini_hyphen_instruct.2fe192450127e6a83f7441aef
[54/54 48:33, Epoch 3/3]
```

Step	Training Loss
10	11.623083
20	11.206940
30	10.010300
40	8.632491
50	7.047097

```
TrainOutput(global_step=54, training_loss=9.438973320855034, metrics={'train_runtime': 2973.5707,
'train_samples_per_second': 0.139, 'train_steps_per_second': 0.018, 'total_flos': 6009742924922880.0,
'train_loss': 9.438973320855034})
```

Figure 4.5: Output

4.4 Making An API For the Model Using Hugging Face

```python
from huggingface_hub import HfApi

api = HfApi()

api.upload_folder(
    folder_path="./nexus-teacher-final",
    repo_id="Diwkarsrd/ai-teacher",
    repo_type="model",
    token="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"
)
```

Figure 4.6: Creating model API

We upload our custom tuned model to Hugging Face Hub where we can access the model using an API, making it very easy to use.

4.5 Project Structure

```
nexus-hosted/
├── frontend/
│   ├── src/App.jsx
│   │
│   ├── public/favicon.svg
│   ├── index.html
│   ├── vite.config.js
│   └── package.json
│
├── backend/
│   ├── src/index.js
│   │
│   │
│   ├── package.json
│   └── .env.example
│
├── ml/
│   ├── training/
│   │   └── train.py
│   │
│   └── data/
│       └── dataset.jsonl
│
└── deploy/
    └── deploy.sh
```

Figure 4.7: Project Structure

5 Testing and Evaluation

5.1 Prototype Implementation

The basic implementation of the AI-Based Virtual Teaching Assistant prototype was performed successfully. The basic functionality which has been included into the system includes:

- Audio recognition as one of the input modes.
- The ability to understand natural language used in questions of the users.
- Creation of responses using the large language model.
- Converting text into voice to deliver responses in spoken format.
- Image generation to create images that explain the response.
- Conversational context handling.
- Avatar interface.

The app runs online and allows the user to communicate with the app in the form of speech or text by asking a question and receiving an answer via multiple interfaces such as audio, text, or even images. The avatar itself moves just lips during the dialogue.

5.2 Functional Testing and Observed Behavior

The prototype underwent testing through scenarios comprising a series of 20 educational questions concerning various scientific facts, mathematics, or computing concepts.

5.2.1 Behavior Observed

When faced with simple factual questions such as "What is the capital of France?" or "Define photosynthesis?" the system produced the correct answer in text form as well as spoke the same aloud. However, in case of voice recognition, the prototype did not respond to questions containing ambient sounds. Conceptual queries requiring an explanation led to satisfactory explanations in paragraph form and even context in case of additional questions such as

"give me an example" or "can you explain that more clearly?" Queries requiring a visual representation led to the successful creation of diagrams representing the concepts asked about. However, once again, the efficiency of this process depended on how difficult the concept was. The system included a rudimentary avatar that provided simultaneous speech to the text displayed; however, its behavior remained unchanged regardless of the type of question posed.

5.3 Identified Limitations

As a whole, the entire system depends extensively on calling API services for recognizing speech, creating a language model, and generating images. Consequently, the response time varied significantly based on internet connectivity speed and load on the API services.

Sometimes the generated image had nothing to do with the user's input, particularly in the case of abstract or unique words. For example, if the user entered "diagram of back-propagation algorithm," the image generated was of some random mathematical formulas.

The lip sync of the virtual assistant did not always match with the generation of speech, which made the process unnatural. Furthermore, the virtual assistant does not have the capability to express emotions through facial expressions based on the context of the conversation.

No functionality is available for creating persistent user profiles and adjusting to their learning experience during the sessions. Every new session starts with zero knowledge about the user's previous experiences and accomplishments. If a question requires intricate reasoning or solving a problem through multiple steps, the answer might not be entirely accurate or complete.

5.4 Summary of Findings

Such an application can be used as a proof of concept that proves that a number of AI-based technologies can be used together to create an AI tutor (speech recognition, speech synthesis, natural language processing and understanding, natural language generation, and avatar animation). The separate components work very effectively in isolated tests, but the overall sequence works less effectively compared to each separate component but creates an

adequate response to the user's query

However, it cannot be considered a working model since it requires some modifications to provide faster responses, generate images, create avatars, and offer personalized responses to the user's questions.

6 Conclusion

The aforementioned report elaborated on the architecture of an AI-powered virtual assistant for teaching applications that make use of the potential of speech recognition, natural language processing, text-to-speech conversion, image generation, and avatar modeling. A working model of the system was constructed demonstrating the technical feasibility of the proposed approach.

Moreover, it was evaluated concerning its functionality and proven to have the ability to recognize the voice/text commands, keep a track record of the conversation, provide corresponding responses, convert them into a speech form, and create pictures, where needed. Avatar illustrates the presence, which operates along with speech synthesis.

However, there are a lot of limitations, such as dependency on APIs, inability to create pictures, basic avatar animations, lack of personalization capabilities, and, finally, no users' evaluations were conducted yet. Consequently, this system can only be used as a proof of concept, but not as an educational software product.

Nonetheless, successful integration of different Artificial Intelligence techniques into one system aimed at providing assistance to teachers should be regarded as a step toward enhancing interaction and engagement in e-learning. In the nearest future, work will be done on testing, modeling, avatar creation, and customization capabilities implementation. Taking these steps into consideration, the proposed solution possesses all necessary prerequisites for bringing closer two distinctively different approaches to learning

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What does 'qualifying text' mean?
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Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

An AI-Powered Embodied Virtual Teaching Assistant for Immersive and Interactive Digital Learning.

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ABSTRACT

Rapid development in AI technologies has enabled the emergence of different approaches to the creation of intelligent learning environments. Regrettably, many e-learning platforms available today lack the necessary level of interactivity, provide no personalized assistance to the learner, and fail to engage the audience efficiently. This paper outlines the process of designing a virtual tutor or conversational agent capable of interacting via speech recognition and text input and applying AI-based approaches for text, voice, and image generation powered by cloud services. In this regard, the suggested framework depends on the use of AI-based services offered by cloud service providers to ensure multilateral learning through text, voice, and image generation. A prototype of the virtual tutor has been designed and tested for functionality. As shown during testing, the proposed system is capable of processing requests either by voice or text, keeping record of the context, generating responses both by voice and text, as well as creating images according to user requests. To coordinate visual and audio feedbacks, an avatar with an authentic appearance is used. Despite being technologically viable, the discussed approach requires further empirical evaluation and personalization.

1. Introduction

A great deal of transformation has taken place owing to rapid advancement in technologies related to AI in different fields such as health, financial services, and education. Particularly, one field which has gone under dramatic transformation because of internet, use of smart devices, and availability of different types of websites is education [1]. Thanks to the aforementioned technological innovations, now it is possible for students from all backgrounds to easily access learning resources. However, there exist different challenges in these learning platforms.

Education within classrooms provides an effective mode of communication where teachers get to communicate directly with their students. This allows them to analyze the academic performance of students, identify learning deficiencies, and find ways to solve these problems. The interactive approach to learning within classroom education helps students learn more effectively. Contrary to traditional education, many of today's technological systems adopt non-interactive means including recording classes using videos and other methods [9].

This is because the current digital learning environment falls short in terms of context and interaction. The student finds it hard when trying to solve queries or exploring complex topics, since most platforms do not facilitate real-time interaction with intelligent systems.

While some of the systems have integrated a chatbot to assist in providing answers to specific questions, the chatbot is unable to answer any complex question due to its inability to understand a complicated query in its entirety [4]. Therefore, the learners may be provided with vague and irrelevant responses that could affect their educational process in a negative way.

Personalized teaching in online learning environments represents another crucial issue to be considered in the learning process. Individuals have different ways of learning and thinking. They come up with a different level of pre-knowledge. Yet many of the current systems utilize one standard way of distribution of learning materials without considering individuals. Therefore, poor outcomes are experienced by those students who require personalized teaching.

One of such approaches that can address the challenges mentioned above is multimodal learning. Multimodal learning refers to the use of various types of information to improve understanding. In the theory of multimedia learning, multimodal learning is expected to enhance understanding through the use of multiple ways to convey information [9]. Visual information, such as graphs and pictures, allows learners to understand a particular concept easily. The same effect can be achieved through the use of audiovisual information.

However, despite all these advantages, the currently available platforms lack proper utilization of their multimodal properties, leading to an inadequate learning experience. Recent developments in the field of NLP and DL technology made it feasible to create conversational AI that can interact with users just like a human would. This is because of the sophisticated algorithms applied in these conversational agents, such as transformers, which enable them to understand natural languages. So far, these AI agents have demonstrated their efficiency in various fields, from customer service to virtual assistants. In the context of education, conversational AI can open doors for real-time interaction with the content. However, existing uses of conversational AI focus exclusively on text conversation [3].

Interaction within the system constitutes yet another factor that affects the efficiency of the learning process, and here interaction becomes an important element affecting such interaction significantly. Communication of the people is not only exchange of information but also includes the social aspect of interaction. In face-to-face communication within traditional classrooms, the importance of the teacher can play an important role in motivating the learner. In online classes, however, the social aspect is missing, leading to a lack of engagement of the learners in the learning process itself. This has led scholars to search for a solution through the use of avatar and embodied conversational agents [11].

Apart from interaction and engagement, another crucial requirement in designing an educational system in today's context is scalability and performance. This is due to the fact that as the usage of online systems increases, there arises a need for developing systems that will be able to cope with large amounts of information and respond promptly. The cloud computing technology assumes great importance in achieving this purpose as it facilitates scalability and optimal utilization of resources [4]. With the help of cloud computing technology, it is possible to use sophisticated algorithms to make sure that the system works efficiently.

With the development of technology used in the generative AI, new opportunities appear in relation to intelligent education system advancement. On the one hand, the technology enables the generation of realistic text, images, and even speech. This, in turn, gives an opportunity to speak about dynamic educational content, including videos generated with the help of both image generation models and natural language processing [6][3]. On the other hand, the use of an image generation model might prove to be quite useful in illustrating certain concepts. Lastly, with speech synthesis, the technology enables dialogues.

While much progress has been achieved in this field, the adoption of various types of artificial intelligence (AI) in learning activities as an integrated system seems to be quite difficult. Despite the presence of additional functions provided by the modern technology such as the possibility to engage in chatting and the ability to receive audiovisual materials, no integrated system has been developed yet which will incorporate all elements ensuring efficient learning process.

As for addressing the issues mentioned above, it is proposed here to develop the concept of a virtual teacher assistant that would be based on AI and integrate natural language processing, speech recognition, generative AI and avatar generation technologies. It would help the learner to obtain the required feedback from the tutor and use various types of information presentation channels including text, voice, and graphics-based media.

The proposed system architecture is scalable and flexible and, therefore, allows incorporating cloud computing capabilities so as to achieve real-time operations and AI integration effectively [4]. As a result, a large number of users can easily fit into this system without reducing its effectiveness. Besides, the mentioned flexibility opens the way for further development of this system.

The primary objective of the study is to develop a new type of educational system, known as an intelligent education system, which can overcome the gap that exists between the learning systems based on the classroom and those based on computers. With the use of AI and multimodal interaction in the proposed intelligent education system, the interactivity of the system can be improved.

Key contributions made by the current research paper are described below:

- Use of an AI-based embodied virtual teaching assistant for digital learning has been proposed by the researcher.
- An effective multimodal learning environment combining text and verbal conversations with visual content creation using artificial intelligence has been developed to improve the learning process and enhance concept comprehension.
- The proposed system utilizes avatar-based interface that mimics the movements made by the teacher.
- An intelligent response creation method has been devised to create context-aware responses according to the user inputs.
- A cloud-based system architecture has been considered for the implementation of the proposed system for improved performance and scalability.
- A mathematical model for functioning of the proposed system has been designed which illustrates how responses are generated and learning takes place.
- Functioning of the proposed system has been demonstrated using a prototype and testing.
- The proposed system has been compared with numerous other conventional systems including chatbots and e-learning systems.

2. The Related Work

The development of Artificial Intelligence at a very fast pace has contributed significantly to the creation of intelligent education systems. Advanced intelligent education systems are developed to create a greater level of accessibility, automation, and interaction through technology [10].

Rapid developments in Artificial Intelligence have resulted in advancements in learning systems that can be categorized into chatbot learning systems, intelligent tutoring systems, and multimodal learning systems. The main reason behind the development of these learning systems is their ability to increase access and automate knowledge and interact with learners in an effective manner [10]. However, there are many learning systems which do not possess multimodal, contextual, and humanlike interaction capabilities.

2.1 Intelligent Tutoring Systems

Intelligent Tutoring System (ITS) is one of the application areas where the use of AI can be found in education domains for providing personalizing education in relation to the performance of the individual. This type of system operates like a one-to-one tutor, analyzing the answers given by the student and giving suggestions accordingly [2]. The learning process in the ITS system can be described as follows:
The mathematical formulation of adaptation for learning is:

$$Kt+1 = Kt + \Delta K$$

Here Kt represents the present knowledge state,
and ΔK represents the increment in knowledge state.
The main limitation of traditional ITS is that they have been confined with certain set of predefined rules, and as a result they cannot accommodate open-ended problems. [10].

2.2 Conversational AI and Natural Language Processing

With the progress made in NLP, there is now the possibility of building conversational AI that can interact with users in a manner similar to how humans interact using approaches such as tokenization, semantic understanding, and context embedding [4].
Through recent advancements in transformer-based models, the creation of responses based on contextual information is now possible. The formulation for generating response probabilities is expressed as:

$$P(R \mid Q) = \prod_{i=1}^{n} P\,(r_i \mid r_1, r_2, \ldots, r_{i-1}, Q)$$

where Q denotes the query input, and R is the sequence of responses produced.
Even though these artificial intelligence systems are interactive in character, they still have many limitations that include limiting the transfer of data to text communication only.

2.3 Multimedia and Multimodal Learning Systems

In a multimedia learning environment, multiple modes of information transmission are employed to facilitate learning processes. According to multimedia learning theory, the use of various stimuli modes improves cognition and learning processes [9]. The effectiveness of multimedia learning is expressed through the equation below:

$$L = f\,(T, A, V)$$

where T, A, and V denote input from texts, sounds, and images respectively

Although possessing many positive features, most of these systems cannot facilitate real-time interaction or dynamic content creation [20].

2.4 Avatar-Based and Embodied Learning Systems

Avatar-based systems use human-like interfaces in digital education environments, enhancing engagement and interactions. Avatar-based system mimics actual teaching actions by using a virtual entity that is capable of communicating both verbally and non-verbally [11].
The effectiveness of interaction in an avatar-based system can be expressed as:

$$E = \alpha C + \beta I$$

where C denotes the quality of communication and I represents the level of interaction.
The presence of these pedagogical agents has been shown to significantly increase the motivational and interactive components of learning, a phenomenon called the "persona effect" [19]. Sadly, modern avatar systems do not make use of intelligent AI technology.

2.5 Generative AI in Educational Systems

With the advent of AI technology, there have been several technological advancements that have led to the creation of dynamic content. These include techniques like GANs and transformers, and the process of generating content creates educational content that is highly realistic [6], [3].
Content generation involves:

$$X_{gen} = G(z)$$

where z refers to the hidden variable, and G refers to the generation function. These methods may be used in the creation of visual and interactive material, which would promote better understanding. However, their use in educational software systems remains limited.

2.6 Speech Processing and Voice-Based Interaction

Speech Recognition and Synthesis make it easy for users to interact with computers by speech. It is possible to use technologies of speech recognition and synthesis to translate a speech signal into text or vice versa [7]. The following picture illustrates translation of speech into text:

$$T = f(S)$$

Where S represents the speech signal and T refers to the text output.

Although voice technology increases accessibility, it does not always become part of a complete learning system, but rather works as a stand-alone component.

2.7 Limitations of Existing Systems

However, there are a few limitations associated with the existing methods. Chatbot methodology uses interaction without multimodality, multimedia methodology offers better presentations but does not provide any interactivity, while the avatar methodology offers interaction without intelligence and context. This demonstrates that there is a requirement for an integrated system that combines these three features together.

2.8 Summary and Research Gap

To conclude, the above-mentioned research studies have been very effective in studying different components in digital education such as tutors, intelligent agents, and multimodal learning. Undoubtedly, at present, there is no existence of any system where all these components will be merged together.
The proposed system is developed based on natural language processing, speech interface, generative artificial intelligence, and avatar interface technologies. It has been developed to address any problems faced by the previous studies [1], [4], [9].

3. The Proposed System / Algorithm

The Proposed System

In this paper, we will attempt to improve the capabilities of the digital learning system by highlighting the limitations of current solutions that utilize chatbots and e-learning programs, such as low interactivity, context understanding, and multimodality. In order to solve the above problems, we will offer a solution that will leverage AI technology to develop an embodied virtual assistant for education.

Our proposed solution will feature several modules of artificial intelligence for adaptive learning. In addition to traditional methods used in other solutions, such as the employment of a single communication channel, our solution will incorporate multimodal communication to maximize the learning process. Our solution is designed to respond adequately to context, explain concepts visually, and mimic human behavior in education through the use of avatars. The overall purpose of our solution is to enhance the learning process and ensure real-time interaction. It can be defined as an optimization task:

$$L_{opt} = \max\,(E + I + A)$$

where E represents engagement, I denotes interactivity, and A indicates adaptability of the system.

3.1 System Modelling and Purpose of Equations

The suggested architecture is to be represented as an interactive system in which the process of processing user inputs consists of several steps, including data collection, natural language processing, answer creation, and output generation.
User input may be viewed as follows:

$$Q = \{q1, q2, \ldots, qn\}$$

Input processing by the proposed system entails employing an NLP model that produces a corresponding response R:

$$R = f(Q, C)$$

where C refers to the contextual data acquired from previous user conversations.
For proper processing and generation of relevant outputs, the following probability function is used for optimization:

$$P(R \mid Q) = \prod_{i=1}^{n} P(ri \mid Q, r1, r2, \ldots, ri-1)$$

3.2 State-Based Interaction Modelling

The process of interaction between user and the system is described by means of state transitions, wherein every interaction becomes a state transition from one state to another.
Let the state of the system be described as:

$$St = (Qt, Ct),$$

where Qt is the query issued by the user at time t and Ct is the context at that time. Next state can be written as:

$$St+1 = St + \Delta S,$$

where ΔS is the change in context and knowledge. In order to enhance interactions, system uses feedback from the user along with past interactions.

3.3 Multimodal Response Generation

One of the most important attributes of the new system under consideration is multimodal response generation, where responses are created using text, speech, and visuals.

Overall responses can be expressed using the formula:

$$O = T + V + A$$

where T refers to text response, V stands for visual content (pictures), and A is an audio response.

3.4 Adaptive Learning Mechanism

To enhance personalization, there is a learning adaptation process that learns how to respond based on interaction.
The formula for this learning adaptation will be:

$$Ut+1 = Ut + \alpha(Ft)$$

where Ut is the knowledge of the user, Ft is the feedback, and α is the learning rate.
This ensures the growth of the system and provides a better response to the users.

3.5 System Workflow and Architecture

The proposed design follows a multi-layer intelligent architecture that consists of several different types of AI technologies that help deliver an interactive and engaging learning experience. As shown in Fig. 1, the proposed solution has the following layers:
User Interaction Layer, Input Processing Layer, AI Intelligence Layer, Multimodal Generation Layer, Avatar Interaction Layer, and Cloud Computing Layer.

The user interaction layer forms the user interface that helps the user interact with the system and submit their input data through both text and voice input modes. The input processing layer helps convert the voice input into text while carrying out various other pre-processing activities like tokenization and normalization.

The intelligence layer forms the core of any AI-powered solution, as it is at this level that natural language processing is carried out in order to comprehend the query submitted by the user and provide a corresponding response. Context management helps the solution remember previous conversations so that they may be taken into account while generating a response.

The multimodal generation layer enhances the learning experience since the generated response can come in several different formats like texts, images, and voice responses. This is achieved using various generative AI models and text-to-speech models.

The avatar interaction layer embodies the human face in the AI-supported system, whereby the responses are generated by virtual human tutors. The lip-syncing and facial expressions of the avatars determine how realistic the behaviors appear.

The output delivery layer marks the stage where the response to the user's question is delivered using a combination of texts, audio, and visual components. Furthermore, the cloud and data layer guarantees the scalability and efficient data management in the system by collecting data on user behavior and combining AI services.

Such an architecture enables real-time query processing, scalability, and integration of different AI services.

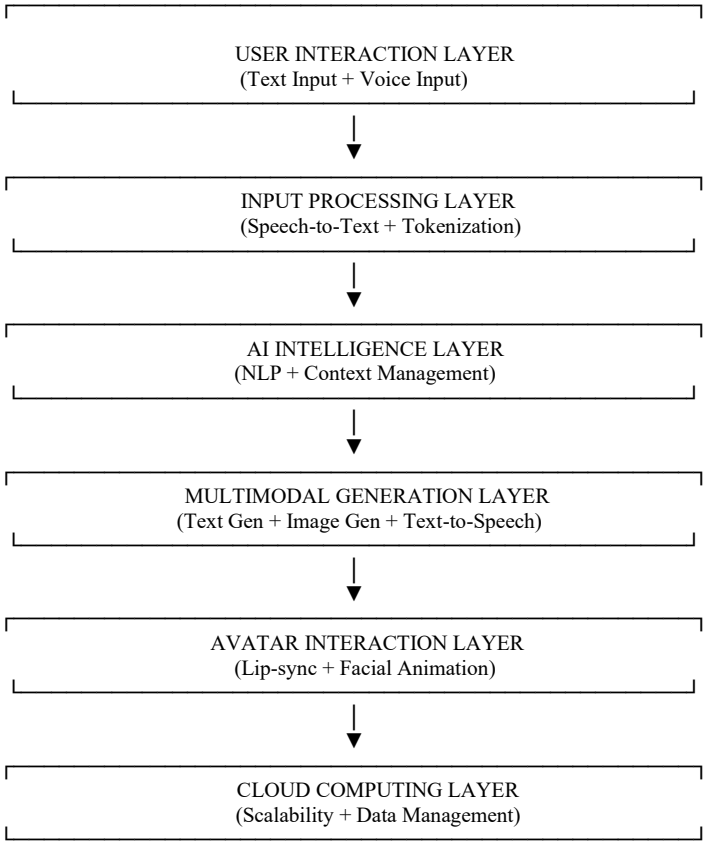

Fig 1: Architecture of the Proposed AI-Based Virtual Teaching System

3.6 Time Complexity Analysis

Time complexity is one factor that will be used to determine the effectiveness of this system design. Here, N represents the number of input data from users, while M represents the complexity of the AI model.
This means that the time complexity equation can be written as follows:

$$O\ (N \times M)$$

Cloud computing allows operations to take place within a negligible amount of time.

3.7 Summary of Proposed System

To sum up, the proposed system has incorporated several forms of AI in a single system, making it possible for learning to be enjoyable and interactive. This is possible due to the incorporation of Natural Language Processing, voice recognition, Generative AI, and avatars in the proposed system, making it possible for the proposed system to overcome limitations found in today's learning systems.
These include modeling, adaptation, and multiple channels.

4. Results and Demonstration

4.1 Prototype Implementation

The basic implementation of the AI-Based Virtual Teaching Assistant prototype was performed successfully. The basic functionality which has been included into the system includes:

- Audio recognition as one of the input modes;
- The ability to understand natural language used in questions of the users;
- Creation of responses using the large language model;
- Converting text into voice to deliver responses in spoken format;
- Image generation to create images that explain the response;
- Conversational context handling;
- Avatar interface.

The app runs online and allows the user to communicate with the app in the form of speech or text by asking a question and receiving an answer via multiple interfaces such as audio, text, or even images. The avatar itself moves just lips during the dialogue.

4.2 Functional Testing and Observed Behavior

The prototype underwent testing through scenarios comprising a series of 20 educational questions concerning various scientific facts, mathematics, or computing concepts.

Behavior Observed:

When faced with simple factual questions such as "What is the capital of France?" or "Define photosynthesis?" the system produced the correct answer in text form as well as spoke the same aloud. However, in case of voice recognition, the prototype did not respond to questions containing ambient sounds.

Conceptual queries requiring an explanation led to satisfactory explanations in paragraph form and even context in case of additional questions such as "give me an example" or "can you explain that more clearly?"

Queries requiring a visual representation led to the successful creation of diagrams representing the concepts asked about. However, once again, the efficiency of this process depended on how difficult the concept was.

The system included a rudimentary avatar that provided simultaneous speech to the text displayed; however, its behavior remained unchanged regardless of the type of question posed.

4.3 Identified Limitations

Here are some of the limitations noticed in the functional testing process:

As a whole, the entire system depends extensively on calling API services for recognizing speech, creating a language model, and generating images. Consequently, the response time varied significantly based on internet connectivity speed and load on the API services.

Sometimes the generated image had nothing to do with the user's input, particularly in the case of abstract or unique words. For example, if the user entered "diagram of backpropagation algorithm," the image generated was of some random mathematical formulas.

The lip sync of the virtual assistant did not always match with the generation of speech, which made the process unnatural. Furthermore, the virtual assistant does not have the capability to express emotions through facial expressions based on the context of the conversation.

No functionality is available for creating persistent user profiles and adjusting to their learning experience during the sessions. Every new session starts with zero knowledge about the user's previous experiences and accomplishments.

If a question requires intricate reasoning or solving a problem through multiple steps, the answer might not be entirely accurate or complete.

4.4 Successful Demonstration Cases

Still, the prototype has managed to convince that there are several aspects that distinguish it from classic e-learning applications:

For example, in one case, after a user had input "What is machine learning?" and then, immediately after, said "Give me an example," the system understood that the latter query was concerned with machine learning and provided an example associated with spam filter technologies. It demonstrates the system's capability to retain context across two consecutive interactions.

Also, in another case, after a user had instructed the system "Draw a binary search tree," a corresponding graphical depiction of nodes appeared as a result of interaction. Though it was not as accurate as required, nonetheless, it communicated the necessary meaning to the user.

Lastly, in the case of a voice test, after a user had issued a command "Define quantum entanglement," a voice recognition process occurred, followed by an interpretation of the query, providing a textual response, which was also read out and displayed visually.

4.5 Summary of Findings

Such an application can be used as a proof of concept that proves that a number of AI-based technologies can be used together to create an AI tutor (speech recognition, speech synthesis, natural language processing and understanding, natural language generation, and avatar animation). The separate components work very effectively in isolated tests, but the overall sequence works less effectively compared to each separate component but creates an adequate response to the user's query.

However, it cannot be considered a working prototype since it requires some modifications to provide faster responses, generate images, create avatars, and offer personalized responses to the user's questions.

4.6 Future Work Directions

With respect to the status and limitations of the existing prototype, the following is anticipated in the future work:

Short Term (over next 2-3 months):

- Conduct user testing involving 10-15 students to gather their feedback concerning usability and relevance
- Implements response caching system for questions that are often asked in order to reduce lag time
- Lip-syncing capability of the avatar through browser-based Web Speech API

Medium Term (over next 3-6 months):

- Trying out the product in real-world settings such as in the class environment for one week
- Logging and analysis of logs to determine common failure points
- Basic personalized features such as storing user names and preferred topic of discussion

Long Term (over next 6-12 months):

- Substitute all cloud-based APIs with local implementation of opensource models such as Whisper (speech), Llama 3 (texts) and Stable Diffusion (images) in order to get rid of any external dependence
- Evaluate the effectiveness of the product versus the traditional chatbot in a laboratory-like setting for competitive edge in learning and interaction
- Use voice emotion detection technology to modify teaching style

4.7 Conclusion

The aforementioned research paper elaborated on the architecture of an AI-powered virtual assistant for teaching applications that make use of the potential of speech recognition, natural language processing, text-to-speech conversion, image generation, and avatar modeling. A working model of the system was constructed demonstrating the technical feasibility of the proposed approach..

Moreover, it was evaluated concerning its functionality and proven to have the ability to recognize the voice/text commands, keep a track record of the conversation, provide corresponding responses, convert them into a speech form, and create pictures, where needed. Avatar illustrates the presence, which operates along with speech synthesis.

However, there are a lot of limitations, such as dependency on APIs, inability to create pictures, basic avatar animations, lack of personalization capabilities, and, finally, no users' evaluations were conducted yet. Consequently, this system can only be used as a proof of concept, but not as an educational software product.

Nonetheless, successful integration of different Artificial Intelligence techniques into one system aimed at providing assistance to teachers should be regarded as a step toward enhancing interaction and engagement in e-learning. In the nearest future, work will be done on testing, modeling, avatar creation, and customization capabilities implementation. Taking these steps into consideration, the proposed solution possesses all necessary prerequisites for bringing closer two distinctively different approaches to learning

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